Fundamentals 25

Siddharth K

May 2025

Problem 25.1

Since $\bar{A}$ is the intersection of all closed sets containing A and C is a closed set containing A, $\bar{A} \subseteq C$.

Since A° is the union of all open subsets of A and O is an open subset of A, $O \subseteq A^\circ$.

Problem 25.2

Since $\bar{A}$ is closed, it's the smallest closed set containing itself, so $\overline{(\bar{A})} = \bar{A}$. Since A° is open, it's the largest open set contained in itself, so $(A^\circ)^\circ = A^\circ$.

Problem 25.3

Since A is closed in Y, $Y \backslash A$ is open. Also, we have that $X \backslash A = (X \backslash Y) \cup (Y \backslash A)$. Since Y is closed, $X \backslash Y$ is open, so $X \backslash A$ is a union of open sets, which is open, so A is closed in X.

Problem 25.5

Assume $f(X)$ is not connected in Y, then there are open sets $A, B \subseteq f(X)$ with $A \cup B = f(X)$. Now consider $f^{-1}(A \cap B)$. We know that

$$f^{-1}(A) \cap f^{-1}(B) \subseteq f^{-1}(A \cap B) = \varnothing.$$

Since f is continuous, $f^{-1}(A)$, $f^{-1}(B)$, and $f^{-1}(A \cap B)$ are open. Also,

$$f^{-1}(A) \cup f^{-1}(B) \supseteq f^{-1}(A \cup B) = X.$$

So we have split X into two disjoint open sets, so it is disconnected. This is a contradiction, so $f(X)$ must be connected in Y.

Problem 25.7

First assume $S \subseteq \mathbb{R}^2$ is horizontal-vertical path connected. So given any $a, b \in S$, there is a horizontal-vertical path $f : [0,1] \to S$ connecting a to b. Since it is made of piecewise continuous components, it is a path as well, so S is path connected.

Now assume $S \subseteq \mathbb{R}^2$ is path connected and open. Let $a, b \in S$ and let f be a path connecting them. We construct the horizontal-vertical path g as such:

1. Let $x_0 = a$.
2. For all $i \geq 0$, let
$$\varepsilon_i = \sup_{\varepsilon} \{\varepsilon > 0 : B_\varepsilon(x_i) \subseteq S\}$$
(the maximal radius for an open ball centered at x_i contained in S).
3. Let
$$x_{i+1} \in (f([0,1]) \cap \partial B_{\varepsilon_i}(x_i)) \backslash \{y \in B_{\varepsilon_j}(x_j), j \leq i\}$$
(the intersection of the path f and the boundary of the ball $B_{\varepsilon_i}(x_i)$ excluding points that were in any previous ball).
4. If $b \in B_{\varepsilon_i}(x_i)$, let $x_{i+1} = b$ instead and stop. Let $n = i + 1$ at this point.
5. For each $x_i = (x_i^{(0)}, x_i^{(1)})$, let
$$g\left(\frac{i}{n}\right) = x_i, \quad \text{and } g\left(\frac{i+0.5}{n}\right) = (x_{i+1}^{(0)}, x_i^{(1)})$$
(except when $i = n$, don't define $g\left(\frac{i+0.5}{n}\right)$).
6. For all $x \in [0,1)$, define
$$g(x) = (x - y)g(ny) + (y - x)g\left(\frac{2ny+1}{2}\right)$$
where $y = \frac{1}{2n}\lfloor 2nx \rfloor$ (we have that $g(1)$ is defined as b already). This connects horizontal or vertical line segments between the points already defined.

So we have found a continuous function $g : [0,1] \to S$ which is composed of horizontal and vertical segments contained in S.

Problem 25.8

Let $S = \mathbb{R}^4 \backslash \{(x, y, 0, 0) : x, y \in \mathbb{R}\}$ and $a, b \in S$ where $a = (a_1, a_2, a_3, a_4)$ and $b = (b_1, b_2, b_3, b_4)$. Define $f : [0,1] \to S$ as follows:

$$f(0) = a, \quad f(1) = b,$$

$$f(0.5) = \begin{cases} \frac{1}{2}(a+b) & \text{if } a_3+b_3, a_4+b_4 \neq 0, \\ (a_1, a_2, a_3+1, a_4) & \text{if } a_3+b_3 = 0, a_4+b_4 \neq 0, \\ (a_1, a_2, a_3, a_4+1) & \text{if } a_3+b_3 \neq 0, a_4+b_4 = 0, \\ (a_1, a_2, a_3+1, a_4+1) & \text{if } a_3+b_3, a_4+b_4 = 0. \end{cases}$$

$$f(x) = 2xf(n) + (1-2x)f(n+0.5), \quad \text{where } n \in \{0, 0.5\} \text{ and } x \in (n, n+0.5).$$

This function creates either a line segment between a and b or two connected line segments. So f is continuous and by construction is contained in S, so S is path connected.

Problem 25.10

There is a bijection f from S^{n-1} to an $(n-1)$-dimensional hypercube, since we can convert any point in rectangular coordinates in S^{n-1} to hyperspherical coordinates (generalization of polar/spherical coordinates) and drop the radius term, leaving us with $n-1$ elements in $[0, 2\pi)$, which is an $(n-1)$-dimensional hypercube.

Now, let $a, b \in S^{n-1}$. Then define $g : [0,1] \to S^{n-1}$ as

$$g(x) = f^{-1}(xf(a) + (1-x)f(b)).$$

So S^{n-1} is path connected if f^{-1} is continuous.

To show this, let $x = (x_1, \ldots, x_n) \in S^{n-1}$ be equal to $f^{-1}(\theta)$ where $\theta = (\theta_1, \ldots, \theta_{n-1}) \in [0, 2\pi)^{n-1}$. By the formulas for hyperspherical coordinates, each x_i is written as a product of sines and cosines of elements in θ, and since sines and cosines are continuous, f^{-1} is continuous.